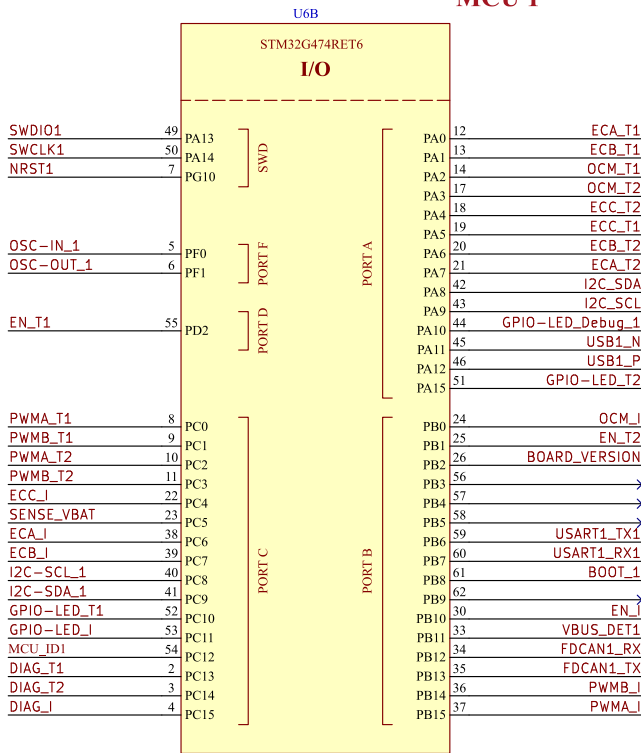
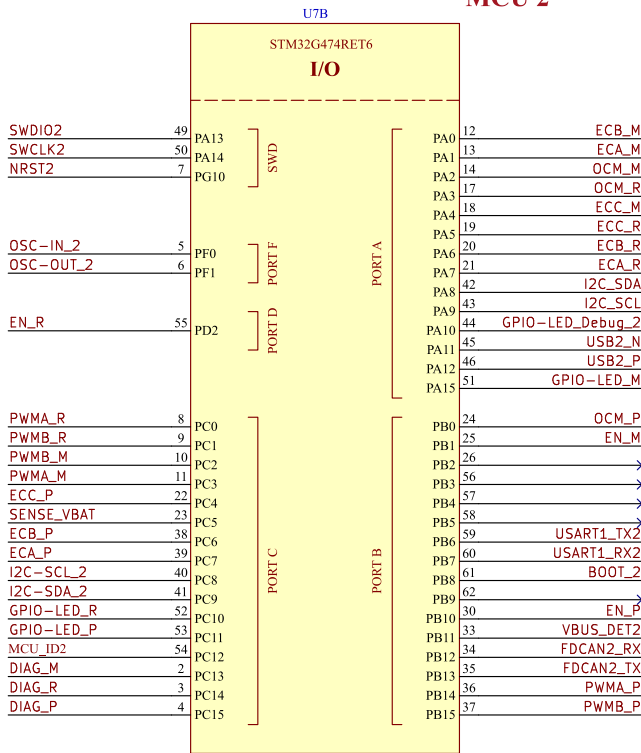


[7] MCU - IOs

MCU 1



MCU 2



SENSE_VBAT

Battery Voltage Sense

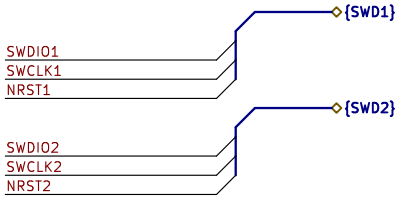
BOARD_VERSION

Board Version Selection

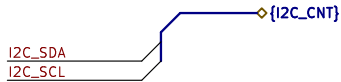
MCB to MB - USART

USART1_RX1
USART1_TX1

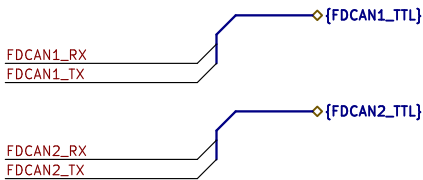
USART1_RX2
USART1_TX2



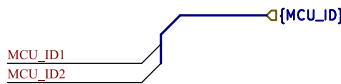
SWD (Debug) Port



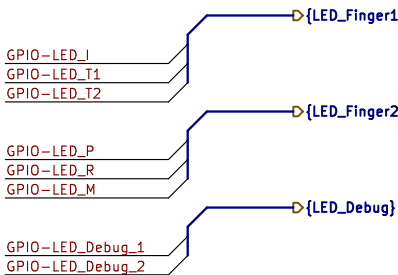
MCB to MB - I2C



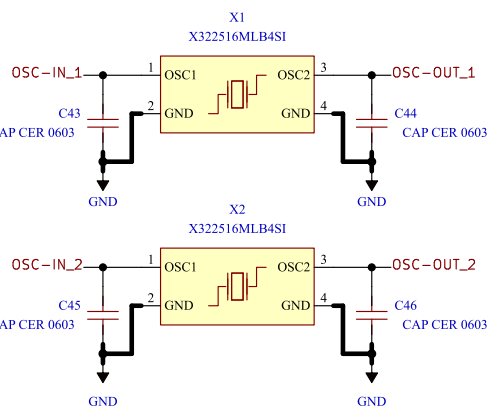
5Mbps FD-CAN Bus



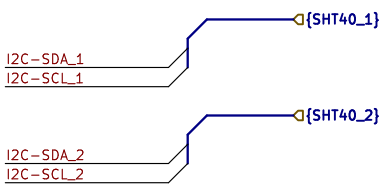
MCU ID



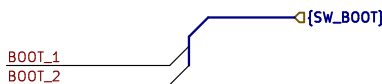
LEDs



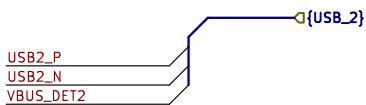
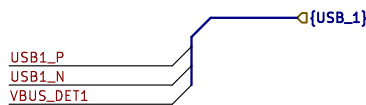
Oscillator HSE



Temperature and Humidity



SWITCH



USB PORT

PWMA_I
PWMB_I
EN_I
DIAG_I
OCM_I
ECA_I
ECB_I
ECC_I

PWMA_M
PWMB_M
EN_M
DIAG_M
OCM_M
ECA_M
ECB_M
ECC_M

PWMA_R
PWMB_R
EN_R
DIAG_R
OCM_R
ECA_R
ECB_R
ECC_R

PWMA_P
PWMB_P
EN_P
DIAG_P
OCM_P
ECA_P
ECB_P
ECC_P

PWMA_T1
PWMB_T1
EN_T1
DIAG_T1
OCM_T1
ECA_T1
ECB_T1
ECC_T1

PWMA_T2
PWMB_T2
EN_T2
DIAG_T2
OCM_T2
ECA_T2
ECB_T2
ECC_T2

Motor Driver

Battery Voltage Sense

SENSE_VBAT

Board Version Selection

BOARD_VERSION

SWD (Debug) Port

NRST2
SWCLK2
SWDIO2

NRST1

SWCLK1

SWDIO1

5Mbps FD-CAN Bus

FDCAN1_TX

FDCAN1_RX

FDCAN2_TX

FDCAN2_RX

LEDs

GPIO-LED_I

GPIO-LED_T1

GPIO-LED_T2

GPIO-LED_P

GPIO-LED_R

GPIO-LED_M

GPIO-LED_Debug_1

GPIO-LED_Debug_2

Temperature and Humidity

I2C-SDA_1

I2C-SCL_1

I2C-SDA_2

I2C-SCL_2

USB PORT

USB1_P

USB1_N

VBUS_DET1

USB2_P

USB2_N

VBUS_DET2

MCB to MB - I2C

I2C_SCL

I2C_SDA

Buzzer

MCU ID

MCU_ID1

MCU_ID2

Test Points

MD Index

PWMA_I

PWMB_I

EN_I

DIAG_I

OCM_I

ECA_I

ECB_I

ECC_I

MD Middle

PWMA_M

PWMB_M

EN_M

DIAG_M

OCM_M

ECA_M

ECB_M

ECC_M

MD Ring

PWMA_R

PWMB_R

EN_R

DIAG_R

OCM_R

ECA_R

ECB_R

ECC_R

MD Pinky

PWMA_P

PWMB_P

EN_P

DIAG_P

OCM_P

ECA_P

ECB_P

ECC_P

MD Thumb 1

PWMA_T1

PWMB_T1

EN_T1

DIAG_T1

OCM_T1

ECA_T1

ECB_T1

ECC_T1

MD Thumb 2

PWMA_T2

PWMB_T2

EN_T2

DIAG_T2

OCM_T2

ECA_T2

ECB_T2

ECC_T2

DESIGN NOTE:
We use the LQFP64
version to have extra
pins for controlling an
external fan.

DESIGN NOTE:
Do not forget to flash
option bytes when
flashing for the first
time.

LAYOUT NOTE:
Footprint deviates from
datasheet and IPC7351 to
allow for soldermask sliver
between pads according to
PCBWay design rules.